

BREATHE-CC · INTEGRATED EXPOSURE INDEX

IEI Monthly Follow-up Additions

Version 4: the form as agreed in the September 28 meeting, with the logic for every question

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Companion files Decision memo, slide deck, BREATHE-CC_DataDictionary_IEI_DEMO.csv (import into a project copy), IEI_new_and_changed_rows.csv, and the interactive preview

In one paragraph

Every addition lives inside the existing monthly follow-up, and no existing question's wording, choices or data changes. A hidden instrument (iei_current_state) keeps each family's latest answers, so in every month after the change goes live a family with nothing new just taps through summaries: 7 taps for a one-home family, 10 to 11 for two homes. The month the change goes live costs 23 taps for one home and 33 to 37 for two, most of them one tap per row in the two routine grids. The medication review (M1, M2) is the one piece that touches an existing section. It is in the proposal for Mari to rule on, and if it stays it removes 3 to 10 medication questions in a no-change month.

What changed after the September 28 meeting

All of it is already in the demo dictionary, the preview and the counts in Section 2.

Item	Raised by	What changed
A3 / A4 columns	Sep 21 and Sep 28	Four columns: Home inside · School or childcare inside · Outside · Part inside, part outside (about half and half). The organized-sports column is gone, since sports counts as Outside, and "camp" is dropped from the school label.
A1 routine grid	Sep 28	Each cell now reads the symbol plus the word (● Home, ■ School, ○ Outside, ● Half and half), so nobody has to look up a key. The Spanish grid carries Spanish words.
C3 / C7 air purifier	Dr. Warden	Cut from the monthly form. Purifier use is asked instead on the 3-month safety perception form (P1), with the other mitigation measures.
C4 / C8 candles, incense	Dr. Warden, Dr. Melaram	Cut. Candle exposure cannot be quantified the way the incense literature does, and incense use here is seasonal at best.
C5 car windows	Dr. Melaram, Dr. Warden	Cut. Little coverage expected in South Texas, and not enough to model against the burden.
D2 / D4 air conditioning	Dr. Warden	Cut. HVAC type is already on file at baseline and predicts use, and parents with smart thermostats cannot answer how often it runs. Windows open (D1, D3) stays.
D10 storms and outages	Dr. Melaram	Kept. Outages are common here, and a generator or a stay somewhere else is a real short-term exposure. Pulling outage data by address is not workable: the Texas grid has many retail providers.
D11 / D12 school change	Dr. Warden	Moved off the monthly form to the 12-month survey only, with the school on file piped in and roughly when the change happened. Published school calendars fill in the rest.

Item	Raised by	What changed
Smoking, monthly review	Dr. Warden	Kept, and repeated monthly for the same reason vaping is. The baseline answer is piped in and the family confirms or updates it. The recall question goes to Driscoll (Section 6).
Vaping items	Dr. Warden	Kept as they were, indexed on days a week and number of people in the home.
M1 / M2 medication review	Dr. Warden	Still in the proposal, flagged ASK MARI, with both objections and the mitigations written out in Section 9.
Counts	Dr. Warden	Reported as taps, split into the month the change goes live and every month after (Section 2).

1. Keeping the monthly burden low

Most of what the IEI needs changes a few times a year, not every month: the child's routine, the schedule between homes, home habits and smoking. So each is asked in full once, saved, and shown back each month for a one-tap check. Families only re-answer what actually changed, and a missed month doesn't erase anything.

The mapped routine instrument: `iei_current_state`

A hidden instrument on the baseline event that families never see, like the geocoding instrument. Each field is a calculated field that holds the latest answer from any monthly survey: it checks Month 18 down to Month 1 and takes the most recent month in which the family entered or changed that section.

Monthly surveys pipe from it. The summary boxes show the saved answers, and choosing "Something changed" opens the questions already filled in with them.

If a family misses a month, nothing is lost. The next survey shows the last saved answers, however long ago they were given.

Each section keeps two dates: when it was last updated and when it was last confirmed.

A staff-only instrument (`iei_staff_followup`) records text or phone follow-up and can correct the custody pattern.

How one family moves through it

Month	What the family sees	What is saved
Month 1	Full questions (routine, homes, habits, smoking and vaping), pre-filled from baseline where baseline asked the same thing (smoking).	Every answer goes into <code>iei_current_state</code> .
Month 2	Summary boxes and "Is this still true?" They tap "Yes, still the same".	Nothing is re-entered; "last confirmed" moves to Month 2.
Month 3	Missed.	Nothing changes; the Month 2 state stands.
Month 4	The same summaries (still from Month 1). They choose "Something changed" for the routine only.	The routine questions open pre-filled; the new answers replace the old routine, and the other sections stay as they were.
School break	Answering "on break" at 0.2 re-opens the routine without a summary, because the saved school-year routine doesn't apply.	The break routine is saved separately by season, so the next school-year month compares against it.

How it is built

- **Edit flag per section.** A hidden calculation (iei_rt_edit, iei_cu_edit, iei_hb_edit, iei_sm_edit, fu_med_edit) is 1 when the family entered or changed that section this month. The state instrument copies a month's answers only when its flag is 1.
- **Frozen snapshot.** Each section stores the saved date as the survey opens (iei_rt_prior and the others). If an older month is reopened after a newer one, its form still reflects what the family saw then, so it can't revert the state or loop.
- **Bilingual summaries.** Multi-Language Management doesn't translate calculated text, so each summary is built twice, in English and Spanish, and the pair is branched on pref_language.
- **Updating.** The state instrument relies on REDCap's cross-event auto-calculation when a monthly survey is saved. If it ever lags, Data Quality Rule H ("incorrect values for calculated fields") recalculates it for all records.

The same idea for the existing medication questions

The monthly form asks the full medication list every month. M1 shows what was reported last time, and M2 asks "Is this still what your child is taking?" "Yes" skips the existing medication questions for the month, and the saved list carries forward. "No" shows them exactly as they are today. For Driscoll this is the clearest sell: families on stable controller medicine stop re-typing drug names every month.

The one change to existing fields

The branching logic of the 10 existing medication fields gets "[fu_med_same]<>'1' and" added in front of what is there now (fu_asthma_meds_types and its five specify boxes, fu_nonpres_meds and its list, fu_pres_meds and its list). Labels, choices, required settings and all saved data stay the same. Months completed before the change still count: the state reads them as full entries.

2. How many taps this adds each month

Counted from the data dictionary itself, by running its branching logic on each scenario. A tap is one answer a parent gives, so each row of a routine grid counts separately; the questions column counts a grid as one. Hidden and display-only fields are not counted.

Scenario	Taps	Questions	Existing medication questions shown
Every month after the change goes live, nothing changed			
One home	7	7	0
Two homes: same nights, or 1st, 3rd and 5th weekends	10	10	0
Two homes: alternating weeks, every other weekend, rotation	11	11	0
Two homes: something else	10	10	0
The month the change goes live, and any later month a family reports a change			
One home	23	11	3
Two homes: same nights every week	37	19	3
Two homes: every other weekend	33	21	3
Two homes: two-week rotation	33	21	3
Two homes: something else	32	20	3
The 12-month survey, where the school question is asked once a year			

Scenario	Taps	Questions	Existing medication questions shown
Nothing changed, school unchanged	8	8	0
School changed	10	10	0
The most it can ever be			
One home: on break, smoking and vaping yes, another start date	27	15	10
Two homes: the same, plus a same-nights pattern and a break stay	48	30	10

For comparison, the current monthly form has 38 answerable fields, 19 of them always shown (23 with the four C-ACT child items). Existing medication questions: 3 always shown, up to 10 with follow-ups.

- **Every month after, one home:** 7 taps. With M2 the medication questions drop by 3 to 10, so the net change runs from +4 to -3.
- **Every month after, two homes:** 10 to 11 taps, with the same medication saving.
- **The month it goes live:** 23 taps for one home, 33 to 37 for two. Fourteen of those are the two routine grids, one tap per time block, and they are pre-filled from then on.
- **Once a year:** the 12-month survey adds the school question, 8 taps in total, or 10 if the school changed.

3. The form, question by question

Each question shows its variable, REDCap type, when it appears (Shown if), which questions it opens or closes (Drives), and why the IEI needs it (Feeds). Flags:

Flag	Meaning
ASK MARI	In the proposal, but it needs Mari's call before submission (Section 9).
CONTINGENCY	Only if Driscoll asks; not in the data dictionary.
DISCUSS	An open format question for this week's meeting.

The section sits after the indoor-air questions and before the hidden QC fields. M1 and M2 sit at the top of the existing Medication Use section.

Section 0. This month

0.1 Contingency only

Use this only if Driscoll is hesitant to add questions. It makes the whole section optional each month ("Would you like to answer the questions about your child's routine and home this month?"). Otherwise it isn't used, and it isn't in the data dictionary.

Why we'd rather not: a skipped month is carried forward from the last answers without a confirmation, so the data gets less certain the more it's used.

0.2 Right now, is your child's school or childcare on its regular school-year schedule?

- Yes, regular school year (1)
- No, on summer break or winter break (2)

- My child doesn't attend school or childcare (3)

Variable	iei_season
Type	Radio, required
Shown if	Always.
Drives	A1, A2, A3, A4, A5, A6, D7
Feeds	Says which routine applies this month (school year or break). It re-opens Section A when the season flips, and a break answer shows the break-stay questions (D7, D8).

0.3 Break reminders (display only)

Summer note: Between May and August, many families' routines change as school lets out or starts back up, including time spent at each home. If your child's days look different now, please choose Something changed below and update them.

Winter note: Around December and January, many families' routines change for winter break, including time spent at each home. If your child's days look different now, please choose Something changed below and update them.

Variable	iei_note_summer, iei_note_winter
Type	Display only
Shown if	The summer note in May to August; the winter note in December and January (from the survey month).
Drives	Nothing below it
Feeds	A reminder at the times routines and custody time usually change, so families update instead of confirming out of habit.

Section A. Daily routine

A1 Routine on file (review box)

What the family sees (sample record)

Your child's routine on file (updated 08/20/2026, last confirmed 09/18/2026)

Time of day	Weekday	Weekend
9 PM to 6 AM (overnight)	● Home	● Home
6 to 8 AM	● Home	● Home
8 to 11 AM	■ School	—
11 AM to 2 PM	■ School	○ Outside
2 to 5 PM	■ School	● Home
5 to 7 PM	⦿ Half and half	—
7 to 9 PM	● Home	● Home

Time in a car or bus: about 30 minutes a day. In place since the start of this school year.

● home, inside ■ school, childcare or camp, inside ○ outside ◇ outside, organized sports ⦿ part inside, part outside

Variable	iei_rt_summary_en, iei_rt_summary_es
Type	Display (piped summary)
Shown if	A routine is on file and this month's school-season answer matches the one on file.
Drives	Nothing below it
Feeds	Shows the saved routine (weekday and weekend grid, car time, start date, last updated, last confirmed) so the family reviews it instead of re-entering it.

A2 Is this still your child's usual routine?

- Yes, still the same (1)
- Something changed, let me update it (2)

Variable	iei_rt_confirm
Type	Radio, required
Shown if	Same as A1.
Drives	A3, A4, A5, A6
Feeds	One tap keeps the routine on file. Something changed opens A3 to A6, pre-filled.

A3 On a typical weekday (Monday to Friday), where is your child for most of each time?

Your best guess is fine; it doesn't have to be exact. If your child is inside somewhere other than school (a relative's home, a store, church), choose Home, inside.

	Home, inside	School or childcare, inside	Outside	Part inside, part outside (about half and half)
9 PM to 6 AM (overnight)	○	○	○	○
6 to 8 AM	○	○	○	○
8 to 11 AM	○	○	○	○
11 AM to 2 PM	○	○	○	○
2 to 5 PM	○	○	○	○
5 to 7 PM	○	○	○	○
7 to 9 PM	○	○	○	○

Variable	iei_rt_wd1 - iei_rt_wd7
Type	Matrix, radio, 7 rows
Shown if	First survey, Something changed at A2, or the season changed. Pre-filled from the file.
Drives	Nothing below it
Feeds	Weekday time-activity weights: the share of each block spent in each microenvironment, the core IEI input. Four columns after the Sep 28 meeting: home inside, school or childcare inside, outside, half and half.
Pre-fill	Pre-filled from the saved answer in iei_current_state (@DEFAULT).

A4 On a typical weekend day (Saturday and Sunday), where is your child for most of each time?

Your best guess is fine; it doesn't have to be exact. If your child is inside somewhere other than school (a relative's home, a store, church), choose Home, inside.

	Home, inside	School or childcare, inside	Outside	Part inside, part outside (about half and half)
9 PM to 6 AM (overnight)	☐	☐	☐	☐
6 to 8 AM	☐	☐	☐	☐
8 to 11 AM	☐	☐	☐	☐
11 AM to 2 PM	☐	☐	☐	☐
2 to 5 PM	☐	☐	☐	☐
5 to 7 PM	☐	☐	☐	☐
7 to 9 PM	☐	☐	☐	☐

Variable	iei_rt_we1 - iei_rt_we7
Type	Matrix, radio, 7 rows
Shown if	Same as A3.
Drives	Nothing below it
Feeds	Weekend time-activity weights. Weekends differ from weekdays in time outdoors and at other homes.
Pre-fill	Pre-filled from the saved answer in iei_current_state (@DEFAULT).

A5 On a typical day, about how long does your child ride in a car, truck or bus? Count all trips together. [DISCUSS](#)

- ☐ Less than 15 minutes (1)
- ☐ About 30 minutes (2)
- ☐ About 1 hour (3)
- ☐ About 2 hours (4)
- ☐ More than 2 hours (5)

Variable	iei_rt_transit
Type	Radio
Shown if	Same as A3.
Drives	Nothing below it
Feeds	In-vehicle time. Minutes in traffic carry high exposure per minute (Jing 2026 splits the day into home and commute).
Discuss	For this week's meeting: switch to hours + minutes boxes and treat car time as a ratio of the day?
Pre-fill	Pre-filled from the saved answer in iei_current_state (@DEFAULT).

A6 Since about when has this been your child's usual routine?

- ☐ Since the start of this school year (1)

- Since school let out for the break (2)
- Only in the past month (3)
- Since my child joined the study ([enrollment date]) (4)
- Another date (after my child joined the study) (5)

About what date?

Please choose a date after your child joined the study.

[date box, M-D-Y]

Variable	iei_rt_since, iei_rt_since_date
Type	Radio
Shown if	Same as A3. The date box appears only for Another date.
Drives	Nothing below it
Feeds	When the routine started, so the exposure record can be back-dated. Answers are capped at the enrollment date (option 4 pipes it in).

Section B. Time between homes (two-home families only)

Section B only appears when baseline multi_household_custody = Yes. Every question in it carries that condition, and D3 to D8 do too.

B1 Schedule between homes on file (review box)

What the family sees (sample record)

Your child's schedule between homes on file (last confirmed 09/18/2026)

The 1st, 3rd and 5th weekends of each month (Texas Standard Possession) (expanded: school pickup Friday to school Monday). In place since joining the study (03/02/2026).

Variable	iei_cu_summary_en, iei_cu_summary_es
Type	Display (piped summary)
Shown if	Two homes at baseline (multi_household_custody = Yes) and a schedule is on file.
Drives	Nothing below it
Feeds	Shows the saved schedule and when it was last confirmed.

B2 Is this still the schedule?

- Yes, still the same (1)
- Something changed, let me update it (2)

Variable	iei_cu_confirm
Type	Radio, required
Shown if	Same as B1.
Drives	B3, B3a, B3b, B4, B5, B6, B7, B8, B9, B11, D5, D5a, D6
Feeds	Keeps the schedule, or opens B3 to B11 pre-filled.

B3 Which best describes how your child's time is split between homes?

Choose the closest one. If none fit, choose Something else and describe it.

- The same nights every week (1)
- One week at each home, switching every week (2)
- Every other weekend at the other home (3)
- The 1st, 3rd and 5th weekends of each month (Texas Standard Possession) (4)
- A two-week rotation, like 2-2-3 or 2-2-5-5 (5)
- Something else, or no set pattern (6)

Variable	iei_cu_pattern
Type	Radio, required
Shown if	Two homes, and first survey or Something changed at B2.
Drives	B3a, B3b, B4, B5, B6, B7, B8, B9, D6
Feeds	Pattern type sets how nights at each home are projected across the month. Every option maps to a projection rule except Something else, which goes to staff follow-up.
Pre-fill	Pre-filled from the saved answer in iei_current_state (@DEFAULT).

Where the six options come from

Option	What it means	Source
1 Same nights every week	The child sleeps at each home on the same nights every week (for example, school nights at one home, Friday and Saturday at the other).	Common fixed schedule (Arizona guide).
2 Alternating weeks (7-7)	A full week at each home, switching on the same day each week.	Arizona Supreme Court, Planning for Parenting Time.
3 Every other weekend	Friday and Saturday nights at the other home every other week, sometimes with one weeknight (B6).	Arizona guide; Custody X Change.
4 1st, 3rd and 5th weekends	Texas Standard Possession Order, for parents within 100 miles: Friday 6 PM to Sunday 6 PM after the 1st, 3rd and 5th Fridays of the month; Thursday 6 to 8 PM during the school year; 30 summer days (July 1 to 31 unless another period is chosen). Expanded: school dismissal to school resumption, which makes the weekend Friday to Monday and Thursday an overnight. Courts must order the expanded times when the parents live within 50 miles, unless the parent declines them or they aren't in the child's best interest.	Tex. Fam. Code §153.312 (standard), §153.317 (expanded times), §153.3171 (within 50 miles).
5 Two-week rotation	2-2-3: 2 nights at one home, 2 at the other, then 3 back at the first; the next week starts at the other home, so each home has 7 of every 14 nights. 2-2-5-5: each home keeps the same 2 weeknights every week (such as Monday–Tuesday and Wednesday–Thursday) and the weekends alternate, which makes 5-night stretches.	Arizona guide; Custody X Change; Novo Law.
6 Something else	Anything else, including no set pattern.	Goes to staff text follow-up (Section 4).

Pared down for fatigue: less common schedules (3-4-4-3, 5-2-2-5, 60/40 splits) fall under "A different two-week rotation" or "Something else" rather than getting their own options. Texas statute defines the standard and expanded weekend times; the other names come from court-published model schedules, since Texas doesn't name them.

B3a When do those weekends usually start and end?

- Friday at school pickup to Monday at school drop-off (1)
- Friday 6 PM to Sunday 6 PM (2)

- Not sure (3)

Variable	iei_cu_expanded
Type	Radio
Shown if	B3 = 1st, 3rd and 5th weekends.
Drives	Nothing below it
Feeds	Texas expanded possession (§153.317) runs the weekend from school dismissal Friday to school Monday: 3 nights instead of 2. Since 2021 courts must order it when the parents live within 50 miles, unless it is declined or not in the child's best interest (§153.3171).
Pre-fill	Pre-filled from the saved answer in iei_current_state (@DEFAULT).

B3b Which rotation is closest?

- 2-2-3: 2 days at one home, 2 at the other, 3 back at the first; the next week flips (1)
- 2-2-5-5: the same 2 weekdays at each home every week, plus alternating 5-day stretches (2)
- A different two-week rotation (3)

Variable	iei_cu_rotation
Type	Radio
Shown if	B3 = two-week rotation.
Drives	Nothing below it
Feeds	Picks the 14-day night sequence (2-2-3 or 2-2-5-5) that is projected from the anchor in B8 and B9.
Pre-fill	Pre-filled from the saved answer in iei_current_state (@DEFAULT).

B4 Please describe the schedule.

Please include: which days or nights your child is at each home; how often it repeats (every week, every 2 weeks, monthly, it changes); and about how many nights a month your child is at the other home.

[notes box]

Variable	iei_cu_describe
Type	Notes box
Shown if	B3 = Something else.
Drives	Nothing below it
Feeds	Free-text description with prompted metrics (days at each home, how often it repeats, nights per month). Sent to staff text follow-up to code a pattern.
Pre-fill	Pre-filled from the saved answer in iei_current_state (@DEFAULT).

B5 Where does your child usually sleep each night?

	Your home	The other home
Sunday night	○	○
Monday night	○	○

	Your home	The other home
Tuesday night	☐	☐
Wednesday night	☐	☐
Thursday night	☐	☐
Friday night	☐	☐
Saturday night	☐	☐

Variable	iei_cu_n_sun - iei_cu_n_sat
Type	Matrix, radio, 7 rows
Shown if	B3 = the same nights every week.
Drives	Nothing below it
Feeds	The weekly night map: nights at each home are projected directly from the day of the week.
Pre-fill	Pre-filled from the saved answer in iei_current_state (@DEFAULT).

B6 Does your child also spend a weeknight at the other home?

A Thursday visit that ends at 8 PM doesn't count; only nights your child sleeps there.

- ☐ No (1)
- ☐ Yes, every week (2)
- ☐ Yes, but only in weeks with a weekend visit (3)

Variable	iei_cu_wknight
Type	Radio
Shown if	B3 = every other weekend, or 1st, 3rd and 5th weekends.
Drives	B7
Feeds	Picks up a weeknight overnight. The Texas standard Thursday visit is 6 to 8 PM, so the answer is No; under expanded possession (§153.317) it runs to school the next morning, so it's an overnight.
Pre-fill	Pre-filled from the saved answer in iei_current_state (@DEFAULT).

B7 Which night?

- ☐ Monday (1)
- ☐ Tuesday (2)
- ☐ Wednesday (3)
- ☐ Thursday (4)

Variable	iei_cu_wknight_day
Type	Radio
Shown if	B6 = Yes (either option).
Drives	Nothing below it
Feeds	The weekday for that overnight.

Pre-fill	Pre-filled from the saved answer in <code>iei_current_state</code> (@DEFAULT).
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B8 Where will your child sleep tonight?

- Your home (1)
- The other home (2)

Variable	<code>iei_cu_tonight</code>
Type	Radio
Shown if	B3 = alternating weeks, every other weekend, or two-week rotation.
Drives	Nothing below it
Feeds	Anchor 1: fixes where the rotation stands on the survey date.

B9 How many nights until your child's next switch to the other home, or back?

If they switch tomorrow, enter 1.

[number box]

Variable	<code>iei_cu_to_switch</code>
Type	Number
Shown if	Same as B8.
Drives	Nothing below it
Feeds	Anchor 2: nights until the next switch sets the phase of the rotation. With B8 it defines the initial schedule; each later month is checked against the projection (D5, D5a).

B11 Since about when has this been the schedule?

- Since the start of this school year (1)
- Since school let out for the break (2)
- Only in the past month (3)
- Since my child joined the study ([enrollment date]) (4)
- Another date (after my child joined the study) (5)

About what date?

Please choose a date after your child joined the study.

[date box, M-D-Y]

Variable	<code>iei_cu_since</code> , <code>iei_cu_since_date</code>
Type	Radio
Shown if	Two homes, and first survey or Something changed.
Drives	Nothing below it
Feeds	Start of this schedule (not before enrollment). The schedule is treated as running from here until the next reported change.

Section C. Home habits

C1 Home habits on file: Is this still true?

What the family sees (sample record)

Home habits on file (updated 08/20/2026, last confirmed 09/18/2026)

Your home ([home address]): Cooking on the stove or in the oven: 2 times a day.

The other home: Cooking on the stove or in the oven: A few times a week.

Is this still true?

- ☐ Yes, still the same (1)
- ☐ Something changed, let me update it (2)

Variable	iei_hb_summary_en, iei_hb_summary_es, iei_hb_confirm
Type	Display (piped summary) + Radio, required
Shown if	Habits are on file.
Drives	C2, C6
Feeds	Review box for both homes with piped addresses. One tap keeps the cooking answers on file.

C2 How often is the stove or oven used to cook at this home?

Answering about your home: [home address]

- ☐ 3 or more times a day (1)
- ☐ 2 times a day (2)
- ☐ 1 time a day (3)
- ☐ A few times a week (4)
- ☐ Once a week or less (5)

Variable	iei_hb_addr_p, iei_hb_cook
Type	Radio
Shown if	First survey or Something changed at C1.
Drives	Nothing below it
Feeds	Indoor source term: gas or electric cooking frequency (fuel type is already on file from cook_fuel).
Pre-fill	Pre-filled from the saved answer in iei_current_state (@DEFAULT).

C6 How often is the stove or oven used to cook at the other home?

Now about the other home: [other home address]

- ☐ 3 or more times a day (1)
- ☐ 2 times a day (2)
- ☐ 1 time a day (3)
- ☐ A few times a week (4)
- ☐ Once a week or less (5)

Variable	iei_hb_addr_s, iei_hb_s_cook
Type	Radio
Shown if	Two homes, and first survey or Something changed.
Drives	Nothing below it
Feeds	Parity: the same indoor source term for the other home, weighted by nights there.
Pre-fill	Pre-filled from the saved answer in iei_current_state (@DEFAULT).

Smoking and vaping (end of Section C)

Baseline asked about smoking once (smoking_in_home, cigarettes_per_day_home, smoking_in_car and the sec_versions) and never asked about vaping. This block turns those into a monthly review. The first monthly survey pre-fills the baseline answers, and later months show the summary.

C9 Smoking and vaping on file: Is this still true?

What the family sees (sample record)

Smoking and vaping on file (as of 08/20/2026)

Your home: Smoking inside: no. Vaping inside: yes, 1 to 2 days a week, One person. Smoking in the car: No, never.

The other home: Smoking inside: yes, about 5 cigarettes a day. Vaping inside: no. Smoking in the car: Yes, occasionally (less than once a week).

Is this still true?

- Yes, still the same (1)
- Something changed, let me update it (2)

Variable	iei_sm_summary_en, iei_sm_summary_es, iei_sm_confirm
Type	Display (piped summary) + Radio, required
Shown if	Smoking and vaping answers are on file from an earlier monthly survey.
Drives	C10, C11, C12, C13, C14, C15, C16, C17, C18, C19, C20, C21
Feeds	Baseline asked about smoking once. This box turns it into a monthly review, so quitting, starting or a new household member is caught.

C10 Does anyone smoke inside this home?

Answering about your home: [home address]

- Yes (1)
- No (0)

Variable	iei_sm_addr_p, iei_sm_home
Type	Radio
Shown if	First monthly survey (pre-filled from baseline smoking_in_home) or Something changed at C9.
Drives	C11
Feeds	Secondhand smoke at home: the largest avoidable indoor PM2.5 source.

Pre-fill	Pre-filled from the saved answer in <code>iei_current_state</code> (@DEFAULT).
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C11 About how many cigarettes are smoked inside this home per day?

[number box]

Variable	<code>iei_sm_cigs</code>
Type	Number
Shown if	C10 = Yes.
Drives	Nothing below it
Feeds	Dose: cigarettes per day, the same scale as baseline <code>cigarettes_per_day_home</code> .
Pre-fill	Pre-filled from the saved answer in <code>iei_current_state</code> (@DEFAULT).

C12 Does anyone smoke in the car when your child is riding in it?

- Child does not ride in cars (1)
- No, never (2)
- Yes, occasionally (less than once a week) (3)
- Yes, regularly (once a week or more) (4)
- Not sure (5)

Variable	<code>iei_sm_car</code>
Type	Radio
Shown if	Same as C10.
Drives	Nothing below it
Feeds	Smoking in the car while the child rides; the same choices as baseline <code>smoking_in_car</code> .
Pre-fill	Pre-filled from the saved answer in <code>iei_current_state</code> (@DEFAULT).

C13 Does anyone use e-cigarettes or vape inside this home?

- Yes (1)
- No (0)

Variable	<code>iei_sm_vape</code>
Type	Radio
Shown if	Same as C10.
Drives	C14, C15
Feeds	Secondhand aerosol. Baseline never asked about vaping.
Pre-fill	Pre-filled from the saved answer in <code>iei_current_state</code> (@DEFAULT).

C14 In a typical week, on how many days does someone vape inside this home?

- Less than once a week (1)

- 1 to 2 days a week (2)
- 3 to 5 days a week (3)
- Every day or nearly every day (4)

Variable	iei_sm_vape_days
Type	Radio
Shown if	C13 = Yes.
Drives	Nothing below it
Feeds	Dose frequency: days a week with indoor vaping, matching the past-7-day indoor exposure measure (Tattan-Birch 2024).
Pre-fill	Pre-filled from the saved answer in iei_current_state (@DEFAULT).

C15 How many people vape inside this home?

- One person (1)
- Two people (2)
- Three or more (3)

Variable	iei_sm_vape_pp1
Type	Radio
Shown if	C13 = Yes.
Drives	Nothing below it
Feeds	Dose intensity: the number of household users, the dose measure used by Tattan-Birch 2024 and Costantino 2024.
Pre-fill	Pre-filled from the saved answer in iei_current_state (@DEFAULT).

C16 Does anyone smoke inside the other home?

Now about the other home: [other home address]

- Yes (1)
- No (0)
- Not sure (99)

Variable	iei_sm_addr_s, iei_sm_s_home
Type	Radio
Shown if	Two homes, and first survey or Something changed.
Drives	C17
Feeds	Parity: smoking at the other home (pre-filled from sec_smoking_in_home).
Pre-fill	Pre-filled from the saved answer in iei_current_state (@DEFAULT).

C17 About how many cigarettes are smoked inside the other home per day?

[number box]

Variable	iei_sm_s_cigs
Type	Number
Shown if	C16 = Yes.
Drives	Nothing below it
Feeds	Parity with C11.
Pre-fill	Pre-filled from the saved answer in iei_current_state (@DEFAULT).

C18 Does anyone from the other home smoke in the car when your child is riding in it?

- Child does not ride in cars (1)
- No, never (2)
- Yes, occasionally (less than once a week) (3)
- Yes, regularly (once a week or more) (4)
- Not sure (5)

Variable	iei_sm_s_car
Type	Radio
Shown if	Same as C16.
Drives	Nothing below it
Feeds	Parity with C12.
Pre-fill	Pre-filled from the saved answer in iei_current_state (@DEFAULT).

C19 Does anyone use e-cigarettes or vape inside the other home?

- Yes (1)
- No (0)
- Not sure (99)

Variable	iei_sm_s_vape
Type	Radio
Shown if	Same as C16.
Drives	C20, C21
Feeds	Parity with C13.
Pre-fill	Pre-filled from the saved answer in iei_current_state (@DEFAULT).

C20 In a typical week, on how many days does someone vape inside the other home?

- Less than once a week (1)
- 1 to 2 days a week (2)
- 3 to 5 days a week (3)
- Every day or nearly every day (4)

Variable	iei_sm_s_vape_days
-----------------	--------------------

Type	Radio
Shown if	C19 = Yes.
Drives	Nothing below it
Feeds	Parity with C14.
Pre-fill	Pre-filled from the saved answer in <code>iei_current_state</code> (@DEFAULT).

C21 How many people vape inside the other home?

- One person (1)
- Two people (2)
- Three or more (3)

Variable	<code>iei_sm_s_vape_pp1</code>
Type	Radio
Shown if	C19 = Yes.
Drives	Nothing below it
Feeds	Parity with C15.
Pre-fill	Pre-filled from the saved answer in <code>iei_current_state</code> (@DEFAULT).

Section D. This past month

D1 In the past month, how often were the windows open at this home?

This past month at your home: [home address]

- Rarely or never (1)
- A few hours a week (2)
- A few hours on most days (3)
- Open most of the day (4)

Variable	<code>iei_mo_addr_p</code> , <code>iei_mo_windows</code>
Type	Radio
Shown if	Always.
Drives	Nothing below it
Feeds	Infiltration: how much outdoor air gets inside (open windows raise <code>F_inf</code>).

D3 In the past month, how often were the windows open at the other home?

This past month at the other home: [other home address]

- Rarely or never (1)
- A few hours a week (2)
- A few hours on most days (3)
- Open most of the day (4)
- Not sure (5)

Variable	iei_mo_addr_s, iei_mo_s_windows
Type	Radio
Shown if	Two homes.
Drives	Nothing below it
Feeds	Infiltration at the other home (address piped so the family knows which home).

D5 Where will your child sleep tonight?

- Your home (1)
- The other home (2)
- Somewhere else (3)

Variable	iei_mo_tonight
Type	Radio
Shown if	Two homes, the schedule was just confirmed (not edited), and the pattern on file is 1 to 5.
Drives	Nothing below it
Feeds	QC check: tonight's answer is compared with the projected schedule. A mismatch flags the record for a staff text.

D5a How many nights until your child's next switch to the other home, or back?

If they switch tomorrow, enter 1.

[number box]

Variable	iei_mo_to_switch
Type	Number
Shown if	Same as D5, and the pattern on file is alternating weeks, every other weekend or a two-week rotation.
Drives	Nothing below it
Feeds	QC check on the rotation's phase. A mismatch flags the record for a staff text; a confirmed answer re-anchors the projection.

D6 In the past month, about how much time was your child at the other home?

- None (1)
- A few nights (2)
- About a week (3)
- About two weeks (4)
- More than half the month (5)

Variable	iei_mo_share
Type	Radio
Shown if	Two homes and the pattern is Something else (on file or chosen this month).
Drives	Nothing below it

Feeds	A rough monthly share of time at the other home when no pattern can be projected.
--------------	---

D7 During this break, is your child staying at the other home for a week or longer?

- Yes (1)
- No (0)

Variable	iei_mo_break_stay
Type	Yes / No
Shown if	Two homes and 0.2 = on break.
Drives	D8
Feeds	Long break stays break the school-year pattern; this catches them without asking about every night.

D8 About when does that stay start?

[date box, M-D-Y]

About when does it end?

[date box, M-D-Y]

Variable	iei_mo_break_start, iei_mo_break_end
Type	Date (M-D-Y)
Shown if	D7 = Yes.
Drives	Nothing below it
Feeds	Date range the other-home weighting overrides the pattern.

D10 In the past month, did any of these happen?

- ☐ The power was out for 24 hours or more (1)
- ☐ A generator was used at your home (2)
- ☐ You stayed somewhere else because of a storm or power outage (3)
- ☐ None of these (4)

Variable	iei_mo_storm
Type	Checkbox
Shown if	Always.
Drives	Nothing below it
Feeds	Kept at Dr. Melaram's request: Corpus Christi outages are common and a generator or a stay elsewhere is a real short-term exposure that helps explain an off-pattern exacerbation. Outage data can't be pulled by address, because the Texas grid has many retail providers.

D11 In the past year, has your child's school or childcare changed, or will it change for the new school year?*Shown above: School or childcare on file: [school on file] (or "none").*

- Yes (1)
- No (0)

Variable	iei_mo_school_disp, iei_mo_school_none, iei_mo_school_chg
Type	Yes / No
Shown if	The 12-month survey only ([event-name] = month_12_arm_1). The school or childcare on file is piped above the question.
Drives	D12
Feeds	School is fixed at intake and never asked again (for example, preschool to kindergarten), and a change moves the school geocode. Asked once a year rather than monthly, so it costs one question a year per family.

D12 What is the name and city of the new school or childcare?

[text box]

About when did (or will) that change happen?

[date box, M-D-Y]

Variable	iei_mo_school_new, iei_mo_school_when
Type	Text
Shown if	D11 = Yes.
Drives	Nothing below it
Feeds	New school name and city for staff geocoding, plus roughly when it changed so the school location can be dated.

Medication review (inside the existing Medication Use section)**M1 Medicines on file (review box) ASK MARI***What the family sees (sample record)***Medicines on file (last updated 08/20/2026)**

Asthma medicines: Bronchodilators (albuterol); Inhaled corticosteroids (Flovent). Regular non-prescription: Zyrtec.
No other prescriptions.

Variable	fu_med_summary_en, fu_med_summary_es
Type	Display (piped summary)
Shown if	Medicines are on file from an earlier monthly survey.
Drives	Nothing below it
Feeds	Shows last month's asthma medicine types and names, non-prescription and other prescriptions.

M2 Is this still what your child is taking? ASK MARI DISCUSS

- Yes, same as before (1)
- No, something changed (2)

Variable	fu_med_same
Type	Radio, required
Shown if	Same as M1.
Drives	existing medication questions
Feeds	Yes skips the existing medication questions for the month (the state keeps last month's answers). No shows them as they are today.
Discuss	Dr. Warden's two concerns, for Mari: it changes how an existing section is presented mid-study (IRB), and medication reporting may look more complete after the switch than before. Mitigations: flag the switch month in the data, verify against the EHR, and run a before/after sensitivity analysis.

After M2: the existing questions (fu_asthma_meds_types, the five specify boxes, fu_nonpres_meds and list, fu_pres_meds and list) appear only when M2 isn't "Yes, same as before".

4. Checking the schedule between homes

Start and end of each schedule

- A schedule applies from its B11 "since" date (never before enrollment) until the day before the next reported change.
- A break stay (D7, D8) overrides the pattern between its start and end dates.
- A staff correction in iei_staff_followup applies from its contact date until the family next updates Section B.
- Nights count where the child sleeps. A switch day counts as half a day at each home, so there's no handoff-time question.

How nights are projected

Pattern	Nights at the other home	Anchor	Monthly check
1 Same nights	The nights marked in B5, every week.	Not needed (day of week).	D5 tonight
2 Alternating weeks	7 nights on, 7 off.	B8 + B9: the next switch is the survey date + B9 nights.	D5 + D5a
3 Every other weekend	Friday and Saturday nights every 14 days, plus the B6/B7 weeknight.	B8 + B9 fix which weekends.	D5 + D5a
4 1st/3rd/5th weekends	Friday and Saturday after the 1st, 3rd and 5th Friday (Friday to Sunday nights if expanded, B3a), plus the B6/B7 weeknight.	The calendar; no anchor needed.	D5
5 Two-week rotation	The 14-day sequence for 2-2-3 or 2-2-5-5.	B8 + B9 fix where in the sequence the child is.	D5 + D5a
6 Something else	No projection; the D6 monthly share is used.	A staff text codes a pattern.	D6 share

Workflow

1. Initial schedule: the family's B3 answers plus B8 and B9 set the projected calendar from the survey date.
2. Each returning month, the family answers D5 (and D5a for rotating patterns). The analysis script, not REDCap, compares the answers with the projection for that date.
3. A match confirms the schedule, and the date becomes the new anchor.
4. A mismatch, "Somewhere else", or pattern 6 puts the record on a staff text list (a REDCap report filtered on these fields).
5. Staff text the family and log it in `iei_staff_followup`: reason, date, method, whether they were reached, a corrected pattern and a confirmed switch date.
6. A corrected pattern replaces the family's last answer in `iei_current_state` until the family next updates Section B.
7. If it isn't resolved, the pattern's average share is used (for example 4 of 14 nights for every other weekend), and the month is flagged as lower confidence.

Is B9 enough to confirm the schedule?

For patterns 2, 3 and 5, tonight's home (B8) plus nights until the next switch (B9) is enough to fix the phase of a 7- or 14-day cycle. Two anchors in different months over-determine it, which is what lets D5 and D5a catch errors. For pattern 1 the day of the week does the job, and for pattern 4 the calendar does, so those only get the D5 check. Pattern 6 can't be projected, so it always goes to the staff text.

5. Smoking and vaping dose: what the literature measures

Study	What they measured	What it means here
Czogala et al. 2014, Nicotine & Tobacco Research	Chamber study: e-cigarette vapor raised PM2.5 to about 152 $\mu\text{g}/\text{m}^3$ against about 819 for cigarettes (roughly 19%). Nicotine was about 10 times lower, with no CO and no rise in VOCs.	Vaping is a real indoor source but a smaller one, so it gets its own items (C13–C15) instead of being folded into smoking.
Tattan-Birch et al. 2024, JAMA Network Open (NHANES 2017–2020, ages 3–11)	Household users and past-7-day indoor exposure, checked against cotinine: 0.081 $\mu\text{g}/\text{L}$ vapor-only vs 0.494 smoke-exposed vs 0.016 unexposed.	Household questions track a biomarker. C14 (days a week) mirrors the past-7-day framing, and C15 (number of people) mirrors household users.
Costantino et al. 2024 (children with asthma)	E-cigarettes used per day and number of users at home. The link to exacerbations wasn't statistically significant.	Keep vaping as a dose input but weight it cautiously; test it in a sensitivity analysis.

We don't ask puffs or sessions a day: a parent can't report that reliably for other household members. For the IEI, a vaping-only home could start at about 0.2 of a smoking home's indoor PM2.5 term (Czogala) as a sensitivity parameter, not an established weight.

6. Back-dating, recall, and what we ask Driscoll

Routines and schedules

- Families already enrolled when the section launches answer A6 and B11 with a start date. "Since my child joined the study" pipes the enrollment date, and dates before enrollment are rejected.
- School calendars are published, so the start of the school year does not have to come from the family: a routine reported as "since the start of this school year" can be dated from the district calendar for that campus.

- For the thesis itself, nothing is applied before the first IEI survey. Analysis starts in the spring with the monthly follow-ups from launch forward.
- For wider BREATHE-CC use, the team needs a written cutoff for how far back a reported routine may be applied, and every back-dated value carries a flag in the release so anyone can drop them and use only the prospective months.

Smoking: the question for Driscoll

Baseline asked about smoking once, up to six months ago for the earliest participants, and never asked about vaping. The proposal pipes the baseline answer into the first monthly survey and asks the family to confirm or update it, then repeats that check every month.

- **What we ask Driscoll:** is confirming a six-month-old baseline answer acceptable, or should smoking simply start fresh at the first monthly survey and be treated as prospective only?
- **Why not impute the earlier months:** smoking is a count, not a rate that moves smoothly, and people tend to smoke steadily and then stop, rather than drift month to month. Imputing backwards would invent that pattern.
- **What we do either way:** flag the month the change went live for every record, and run a sensitivity analysis comparing the months before and after it, so any jump in reporting is visible rather than buried.

7. How many variables this adds, and the missing-data question

Raised by Dr. Jin: new survey items add missing values, and with a cohort this size a long list of variables strains any model that uses them all.

- **Nothing is analysed one variable at a time.** The routine grid, the schedule between homes and the home-habit items are inputs to two sub-indices (outdoor and indoor), which combine into one exposure value per child per hour. The model sees the index, not the fields.
- **Missingness is structural, not random.** Most blanks come from branching: a one-home family never sees Section B. Those are not missing values, they are questions that do not apply, and they are handled as a separate category rather than imputed.
- **Carry-forward reduces gaps rather than creating them.** A missed month keeps the last saved routine, so a skipped survey does not leave a hole in the exposure series; it lowers confidence for that month, which is recorded.
- **Where the count is still too high,** items are collapsed into constructs before modelling (for example, one indoor-source term from cooking frequency, smoking and vaping), which is the same approach Dr. Jin described.
- **Validation is limited by sample size, and the write-up says so.** The thesis reports the index against the nearest-monitor approach on the cohort we have; a fuller validation, including simulated or external cohort data, belongs to the methods paper later.

8. Proposed additions to the safety perception forms

These go on the existing 3-month primary and secondary household safety perception forms, not the monthly survey. They are in the demo dictionary so the team can see them, marked PROPOSED. P1 is where the air purifier question landed after the September 28 meeting, alongside the other mitigation measures. This also

answers the point Dr. Melaram brought back from an environmental epidemiologist: reviewers ask what families actually do about exposure, and we currently ask nothing.

ID	Question	Choices	Why
P1	Which of these, if any, does your household use to help with your child's asthma or the air at home? (Select all that apply)	Air purifier or air cleaner · Allergen-proof mattress or pillow covers · HEPA vacuum · Dehumidifier or fixing leaks and moisture · Pest control (cockroaches or mice) · Keeping windows closed on bad air days · Keeping my child inside on bad air days · No smoking or vaping inside the home · Other · None of these	Mitigation actually used (mite-proof covers are the strongest evidence, Hogaard 2025; HEPA results are mixed). Can replace the monthly purifier item. Note: owning these tracks income, so read it as an access marker, not only a behavior.
P2	In the past 3 months, how often did you keep your child inside because of bad air, smoke, or a smell from plants or refineries?	Never · Once or twice · Several times · Often · Not sure	Moved from monthly D13: behavior on bad air days, every 3 months.
P3	Where do you usually hear about air quality? (Select all that apply)	A weather or air quality app · Text or email alerts · TV or radio · Social media · My child's school · My child's doctor or clinic · I don't usually check · Other	Where families hear about air quality: tells the team which channel an alert program should use.
P4	Which of these would be most helpful to your family? (Select all that apply)	Text alerts on bad air days · A website or dashboard showing air quality near home and school · Information on reducing asthma triggers at home · Help getting an air purifier or allergen covers · School notices on bad air days · Other · None of these	What would help, including a dashboard of bad air days near home and school.
P5	If you knew in the morning that the air would be bad that day, would it change any decisions about your child's day, like outdoor play or sports?	Yes, definitely · Maybe · No · Not sure	Whether a forecast would change decisions: tests if an alert could change exposure.
P6	At the other home, which of these, if any, are used to help with your child's asthma or the air? (Select all that apply)	Air purifier or air cleaner · Allergen-proof mattress or pillow covers · HEPA vacuum · Dehumidifier or fixing leaks and moisture · Pest control (cockroaches or mice) · Keeping windows closed on bad air days · Keeping my child inside on bad air days · No smoking or vaping inside the home · Other · None of these · Not sure	Parity with P1 for the other home.
P7	In the past 3 months, how often was your child kept inside at the other home because of bad air, smoke, or a smell from plants or refineries?	Never · Once or twice · Several times · Often · Not sure	Parity with P2 for the other home.

Duplicate indoor-air items, not changed

The safety perception forms ask about mold, damp spots and water damage over 3 months, and the monthly form asks the same things every month. Removing or hiding the 3-month versions would mean editing existing fields, so this dictionary leaves them as they are and lists it as an open decision. If the team agrees, @HIDDEN on the three primary and three secondary items would hide them without deleting any data. Deleting fields would delete their data.

Mitigation answers (P1, P6) track household income: covers and purifiers cost money. Read them as an access marker as well as a behavior, and adjust for SES in any analysis.

9. Removed and renumbered items

Old ID	Variable or option	What happened
C3 / C7	iei_hb_purifier, iei_hb_s_purifier	Cut from the monthly form (Sep 28). Purifier use is asked instead on the 3-month safety perception form as part of P1, where it works as a mitigation and access measure.
C4 / C8	iei_hb_burn, iei_hb_s_burn	Cut (Sep 28). Candles and incense can't be quantified here, and the incense literature is mostly from settings unlike Corpus Christi.
C5	iei_hb_car_win	Cut (Sep 28). Almost no coverage expected in South Texas, and the burden isn't worth what it would support.
D2 / D4	iei_mo_ac, iei_mo_s_ac	Cut (Sep 28). HVAC type is already on file at baseline and predicts use; parents with smart thermostats can't answer how often it runs.
A3 / A4 col	Outside, organized sports	Cut as a matrix column (Sep 28). Organized sports counts as Outside.
A3 / A4 col	camp	Dropped from the school column label, which now reads School or childcare, inside.
C4	iei_hb_purifier_run	Removed: how often the purifier runs is unverifiable and cannot be weighted.
B10	iei_cu_handoff	Removed: handoff times are too fine-grained; switch days are treated as half-days (50/50).
D9	iei_mo_sports	Removed: organized sports is now a matrix column in A3 and A4 (Outside, organized sports).
D13	iei_mo_kept_in	Moved to safety perceptions as P2 (iei_p_kept_in), asked every 3 months.
A3/A4 col	Somewhere else inside	Removed as a matrix column; other indoor places count as Home, inside. Replaced by Outside, organized sports.
A3/A4 col	Mostly inside / 75-25	Removed; one Part inside, part outside column that is about half and half.

Items whose ID changed or that are new:

Old	New	Variable
new	B3a	iei_cu_expanded
new	B3b	iei_cu_rotation
new	C6	iei_hb_s_cook
new	C9	iei_sm_summary_en, iei_sm_confirm ...
new	C10	iei_sm_home
new	C11	iei_sm_cigs
new	C12	iei_sm_car
C6	C13	iei_sm_vape
C7	C14	iei_sm_vape_days
C8	C15	iei_sm_vape_ppl
new	C16	iei_sm_s_home
new	C17	iei_sm_s_cigs
new	C18	iei_sm_s_car
C12	C19	iei_sm_s_vape
C13	C20	iei_sm_s_vape_days
C14	C21	iei_sm_s_vape_ppl

Old	New	Variable
new	D5a	iei_mo_to_switch
new	M1	fu_med_summary_en
new	M2	fu_med_same

10. Open decisions, and who decides

The first two need an answer before the IRB submission; the rest can be settled as the dictionary is finalised.

1. M1 and M2, the medication review: Dr. Warden flagged that it changes how an existing section is presented mid-study (IRB) and that medication reporting may look more complete after the switch. It is in the dictionary for now, for Mari to rule on. If she says no, deleting two fields and restoring the old branching removes it.
2. Recall for smoking: baseline smoking is up to six months old for the earliest participants. Ask Driscoll whether a single "is this still true?" against the baseline answer is acceptable, or whether smoking should simply start fresh from the first monthly survey.
3. A5 car time: keep the bands, or hours and minutes boxes treated as a ratio of the day?
4. Safety perceptions: the 3-month mold, damp and water-damage items repeat the monthly indoor-air items. Replacing them with P1 to P5 needs the team's decision, because hiding existing fields is a change to existing fields and deleting them deletes data.
5. Back-dating cutoff: how far before the first IEI survey a reported start date may be applied for BREATHE-CC analyses. The thesis itself uses nothing before the first IEI survey.
6. Matrix format: radio (one place per block, current) or checkbox (all that apply, split evenly)?
7. Dust mites (Dr. Melaram): mite-proof mattress and pillow covers are already an option in P1 on the safety perception form. Is that enough, or should the 3-month form also ask about carpet and bedding directly?
8. Traffic near the home (Dr. Warden's vehicles-per-hour variable): we can derive road class and traffic counts from TxDOT AADT data at the geocoded addresses instead of asking families. Confirm that is the preferred route.

11. Loading the demo into a copy of the project

1. In REDCap, Other Functionality → Copy the project, including all records. The copy starts in Development, so dictionary changes apply immediately.
2. In the copy, download the current data dictionary as a backup.
3. Data Dictionary → upload BREATHE-CC_DataDictionary_IEI_DEMO.csv. The change summary should show 238 new fields, 10 modified fields (branching only) and 2 new instruments (iei_current_state, iei_staff_followup).
4. Project Setup → Designate instruments for my events: tick iei_current_state and iei_staff_followup for baseline_arm_1 only. Don't enable either as a survey.
5. Data Quality → run Rule H and fix all, which fills iei_current_state for existing records (including their past medication answers).

6. Multi-Language Management: the Spanish summaries are already in the dictionary as `_es` fields; the new question labels need the translator's pass.

What to test in the copy

- New test record: the first monthly survey shows every question, and no summaries.
- Submit Month 1, then open Month 2: the summaries show the Month 1 answers, and "Yes, still the same" hides the questions.
- "Something changed" opens the questions pre-filled with the saved answers.
- Skip Month 3: Month 4 still shows the Month 2 state.
- Reopen Month 2 after Month 4 is saved: nothing reverts (frozen snapshot).
- Answer "on break" at 0.2: the routine questions open with no summary, and D7 appears for two-home records.
- A Spanish record (`pref_language = es`) shows the Spanish summaries.
- Two-home record: try each pattern; returning months show D5 (and D5a for patterns 2, 3, 5), or D6 for pattern 6.
- Staff form: a corrected pattern dated after the last monthly survey replaces `iei_st_cu_pattern`, and a later family edit wins again. This tests `datediff` on the saved text dates; if it fails, give the `iei_st_*_asof` fields date validation.
- Existing record with past months: after Rule H, M1 shows the latest medication list, dated "the Month N survey".
- Save a monthly survey and check that `iei_current_state` updates without being opened. If it doesn't, schedule Rule H weekly.
- Take a full first-month survey on a phone and time it.

12. Other issues found while building this

Outside the IEI scope, but worth passing on. None of these are changed in the demo dictionary.

- Spanish address displays translate the piping variable names ("`[dirección_calle]`" in `primary_household_address_display`, and the same in the safety perception address displays), so the addresses don't pipe for Spanish-language families.
- The Spanish label for `smoking_in_home` and `cigarettes_per_day_home` says "nuevo hogar" (new home), carried over from the change-of-address wording.
- `fu_asthma_meds_types` has empty Spanish choice labels.
- `fu_asthma_meds_types` is required but has no "None" option, so a child on no asthma medicine can't be answered honestly. M1 shows "none reported" in that case.

Appendix A. What participants are told before the first changed survey

Sent with the usual invitation for the month the change goes live, in English and Spanish, so nobody opens a longer form without warning. Draft for the team and the translator.

Text message (English)

BREATHE-CC: this month's survey looks a little different. We added a few questions about your child's daily routine and home so we can estimate the air they breathe more accurately. It is longer this one time; after this you will mostly just confirm that nothing has changed. Thank you for staying with us. [survey-link]

Text message (Spanish, draft for the translator)

BREATHE-CC: la encuesta de este mes se ve un poco diferente. Agregamos algunas preguntas sobre la rutina diaria y el hogar de su hijo/a para estimar mejor el aire que respira. Solo esta vez será más larga; después solo confirmará que nada ha cambiado. Gracias por seguir con nosotros. [survey-link]

Email version, additional paragraph

After the first time, most months will show you what we already have on file and ask one question: is this still true? If nothing has changed, you are done in a few taps. If something has changed, you can update just that part. Nothing else about the study, your visits or your compensation changes.

Appendix B. What goes to the IRB

- The new questions, printed as the participant sees them, including the summary boxes with sample piped values (the PDF of Section 3 does this).
- The data dictionary for the copied project, showing the new fields and the branching.
- A note that no existing question changes wording, choices or answer options. The only edit to an existing field is the branching on the 10 medication questions, and only if M1 and M2 are approved.
- The participant notice in Appendix A, in both languages.
- The reason for the change: the additions support an exposure index that uses the monitoring network already in the protocol, and they replace nothing the participants already agreed to.
- A statement on back-dating: reported start dates are never applied before enrollment, and back-dated values are flagged in analysis.

Sources

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